

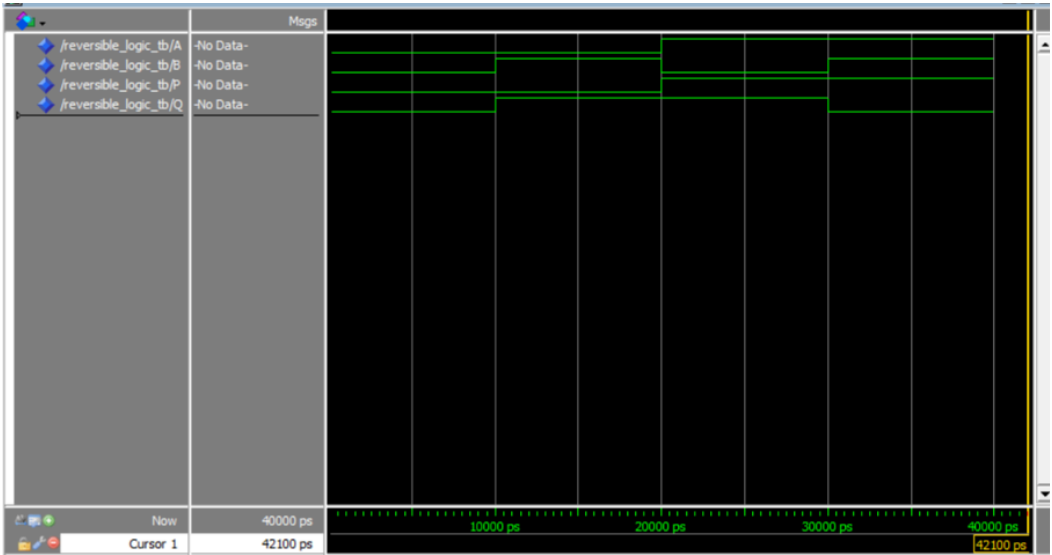
Your testbench should produce these results:

Time	A	B	P	Q
0 ns	0	0	0	0
10 ns	0	1	0	1
20 ns	1	0	1	1
30 ns	1	1	1	0

The outputs follow the reversible Feynman/CNOT gate equations:

- $P = A$
- $Q = A \oplus B$

Modelsim waveform: matches with this



In your reversible-logic project, **P** and **Q** are the two output equations of a Feynman gate (also called a **CNOT gate**).

Your VHDL code defines:

$$P = A$$

$$Q = A \oplus B$$

Here is what each equation means.

1. **P = A — Pass-through output**

- **A** is the control input.
- **P** copies the value of **A** directly.
- If **A** = 0, then **P** = 0.
- If **A** = 1, then **P** = 1.

P preserves the original value of **A**.



2. **Q = A XOR B — XOR output**

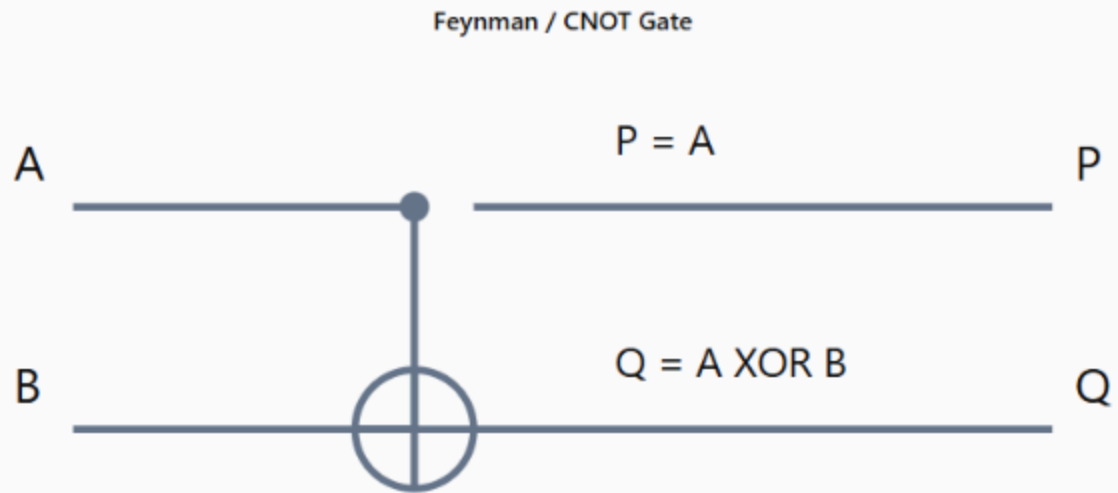
The symbol $\oplus$ means **exclusive OR (XOR)**.

- If **A** and **B** are the same, **Q** = 0.
- If **A** and **B** are different, **Q** = 1.

A	B	P = A	Q = A XOR B
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

The outputs just match whatever place you put the marker on the waveform

3. What does this mean for your FPGA?



- Switch 0 (A) controls the pass-through output P.
- Switch 1 (B) is combined with A using XOR to produce Q.
- LEDR0 displays P.
- LEDR1 displays Q.

Remember: P keeps A unchanged, while Q changes B when A = 1. That is the main operation of the reversible Feynman/CNOT gate.